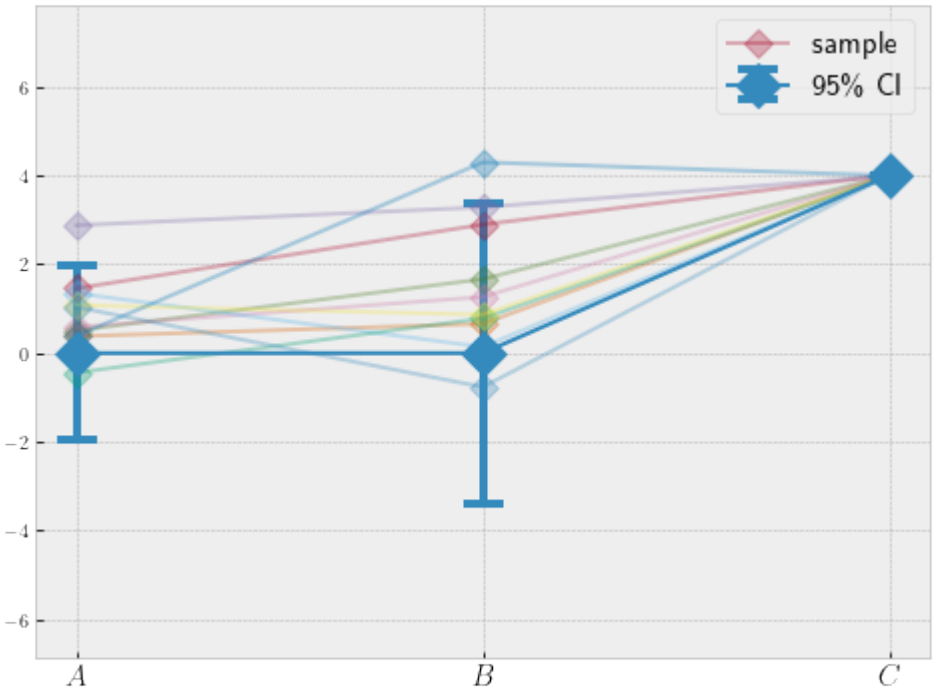


Figure 2.7 Parallel coordinates plot visualizing the multivariate Gaussian from Figure 2.5, conditioned on C = 4. Here, no other marginal distribution changes. All drawn samples interpolate C = 4.

This is when we find out that a house in California was sold. As this house has nothing to do with our own house in Missouri, the belief about the price of our house stays the same.

There is another interesting thing about Figure 2.6 and Figure 2.7. Notice that when we condition on B = 2 in Figure 2.6, all the samples we draw of the posterior multivariate Gaussian pass through the point (B, 2). This is because in our posterior belief, we don't have any uncertainty about what value B takes anymore, and any sample drawn from the posterior distribution needs to satisfy the constraints from this condition. The same thing goes for the point (C, 2) in Figure 2.7.

Visually, you could think of this as when we condition on a variable, we "tie" the samples drawn from the prior distribution (in Figure 2.5) into a knot at the same point corresponding to the variable we condition on, as shown in Figure 2.8.

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